

Scott Phillip Ahrenkiel

Physicist/Programmer

Location: Arvada, Colorado 80005

Phone: 720-988-6627

Email: spahrenk@gmail.com

LinkedIn: www.linkedin.com/in/phil-ahrenkiel-04b5b36

GitHub: github.com/orgs/eQuillTech/repositories

Professional Summary

Ph.D. Physicist with 25+ years of experience combining advanced physics research with software development. Expertise in computational modeling, scientific programming, and algorithm development. Proven track record of developing physics-based simulation libraries, data analysis tools, and visualization applications. Strong background in materials science, electron microscopy, and photovoltaics research with extensive programming experience in C++, Python, and scientific computing frameworks.

Core Technical Skills

Programming Languages: C++, Python, Objective C, Swift, MATLAB, LabVIEW, Igor Pro, Digital Micrograph

Development Tools: VSCode, Xcode, CLion, Git/Mercurial, GitHub, JIRA

Scientific Computing: NumPy, SciPy, Plotly Dash, data visualization, algorithm development

Machine Learning: Completed Stanford/Coursera certifications in Machine Learning, Supervised Learning, Unsupervised Learning

Computational Methods: Numerical simulation, dynamical electron diffraction, image processing, computational physics

Data Formats: JSON, SQL, XML, HDF5

Systems: macOS, Linux, Windows

Domain Expertise: Physics simulation, materials characterization, thermal modeling, energy systems

Professional Experience

Physicist/Programmer

eQuillTech, Arvada, Colorado | *October 2025 - Present*

- Develop physics-based software applications integrating computational modeling with practical engineering applications
- Design and implement algorithms for materials science and physics simulations
- Create open-source scientific computing tools available on GitHub

Computational Scientist

Big Ladder Software, Denver, Colorado | *June 2023 - October 2025*

- **Heat Pump Water Heater Simulation:** Lead developer for HPWHsim library used in California Energy Commission compliance calculations. Restructured legacy codebase with emphasis on physical rigor while maintaining 100+ unit/function tests. Implemented JSON-based data representation for external manufacturer models.
- **Data Visualization & Analysis:** Independently developed interactive graphical analysis suite using Plotly Dash for simulation parameter refinement and manufacturer model validation.
- **Physical Units Library:** Created header-only C++ library for physical-units agnostic API, improving code safety and readability.
- **Energy Resilience Simulation:** Enhanced ERIN library with CLI interface and additional features for catastrophic analysis of large-scale installations (military bases, universities).
- **Data Conversion Tools:** Developed SQL-based data extraction library for legacy Access databases (California Utility Allowance Calculator).
- **Documentation & Testing:** Formalized evaluation methods and generated comprehensive documentation for C++ interpolation library (btwxt).

Software Engineer

Maptek, Golden, Colorado | *April 2022 - February 2023*

- Maintained and enhanced Vulcan mining software using C++ with JIRA-based review system and Mercurial version control
- Developed novel algorithm for generating optimized topographical triangulations with variable resolution from large geo-tagged TIFF files
- Expanded libTIFF driver to handle USGS Global Fiducials Library data
- Created unit tests and fixed defects in production codebase

Professor (Assistant/Associate/Full)

South Dakota School of Mines and Technology, Rapid City, SD | *August 2006 - July 2021*

- **Software Development:** Created Digital Micrograph script manager and plug-ins for JEM-2100 transmission electron microscope, including automation tools for montaging and data archiving
- Developed computational analysis software for precession-diffraction pattern refinement to dynamical simulations
- Built programmable MOCVD control software with extensive hardware customization and safety protocols
- Implemented convergent-beam electron diffraction analysis tools for composition and strain analysis
- **Research Computing:** Applied computational methods to materials characterization including dynamical electron diffraction, image processing, and data analysis
- Developed algorithms for extracting alloy compositions from energy-dispersive X-ray spectrometry data
- **Graduate Education:** Developed and taught graduate courses in nanocharacterization, crystallography, nanoelectronics, and photovoltaics

Senior Scientist I / Scientist II / Postdoctoral Scientist

National Renewable Energy Laboratory (NREL), Golden, Colorado | *January 1995 - July 2006*

- Developed physical models of relaxation behavior in metamorphic semiconductor heterostructures for world-record efficiency photovoltaic devices

- Created theoretical models for TEM image contrast of partially ordered structures and modulated superlattices
 - Developed dynamical electron-diffraction software for computational analysis of ordered semiconductor structures
 - Designed and constructed tunable light source instrumentation for tandem solar cell testing
 - Applied TEM tomography and computational analysis to nanoparticle characterization
-

Education

Ph.D., Physics

University of Colorado, Boulder, Colorado | *Completed December 1995*

Dissertation: Phase Separation and Atomic Ordering in Epitaxial Semiconductor Alloys Studied by Transmission Electron Microscopy

M.S., Physics

Colorado School of Mines, Golden, Colorado | *Completed December 1991*

Thesis: Intercalation Processes in Layered Silicates and High-Tc Superconducting Oxides Studied by X-Ray Diffraction

Developed complete XRD data acquisition/analysis software package

B.S., Engineering Physics

University of Colorado, Boulder, Colorado | *Completed December 1987*

Selected Publications & Patents

Book: Co-author, *Theory and Methods of Photovoltaic Material Characterization*, World Scientific, 2019

Patent Applications:

- Strain balanced direct bandgap aluminum indium phosphide quantum wells for light emitting diodes (US20220271194A1)

- Method for achieving device-quality, lattice-mismatched, heteroepitaxial active layers (US20060048700A1)

Journal Publications: 100+ peer-reviewed publications in physics, materials science, and photovoltaics (Google Scholar profile available)

Grants Awarded

- U.S. Department of Energy: \$1.4M+ in competitive research grants (2007-2018)
 - National Science Foundation: \$699K for TEM acquisition and training (2009)
 - NREL Director's Research Fund: \$120K for nanoparticle characterization (2004)
-

Professional Certifications

- Machine Learning (Stanford/Coursera, 2024)
 - Supervised Machine Learning (Stanford/Coursera, 2024)
 - Unsupervised Learning (Stanford/Coursera, 2024)
-

Professional Memberships

- American Physical Society
 - Microscopy Society of America
 - Materials Research Society
-

Additional Experience

Summer Student, CERN, Geneva, Switzerland | *June-August 1987*

- Installed time-projection chamber for SPS collider experiment
- Procured and installed FASTBUS data-acquisition electronics
- Developed test programs for electronics validation